

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE					
Regular/Supplementary Winter Examination – 2025					
Course: B.Tech		Branch : Computer Engineering		Semester : VI	
Subject Code & Name: BTCOC601_Y23 Compiler Design					
Max Marks: 60		Date:		Duration: 3 Hr.	
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.					
				(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)				12
1	1. Which of the following is NOT a phase of a compiler?				1
	a. Lexical analysis	b. Semantic analysis	c. Memory management	d. Code optimization	
2	In which phase of the compiler is the syntax tree generated?				1
	a. Lexical analysis	b. Syntax analysis	c. Semantic analysis	d. Code generation	
3	Which data structure is typically used to represent the syntax tree during the parsing phase?				1
	a. Linked list	b. Stack	c. Tree	d. Array	
4	Which phase of the compiler is responsible for detecting semantic errors in the program?				1
	a. Lexical analysis	b. Syntax analysis	c. Semantic analysis	d. Code generation	
5	The final phase of a compiler that generates the machine code is:				1
	a. Code generation	b. Code optimization	c. Lexical analysis	d. Semantic analysis	
6	Which of the following is NOT a type of intermediate code representation?				1
	a. Three-address code	b. Abstract syntax tree	c. Assembly code	d. Bytecode	

7	The output of the lexical analysis phase of a compiler is:					1
	a. A syntax tree	b. Tokens	c. Machine code	d. Intermediate code		
8	In the context of compiler construction, LR parsing refers to:					1
	a. A type of lexical analyzer	b. A method of parsing using a left-to-right scan and a rightmost derivation	c. A type of intermediate code	d. A method for code optimization		
9	The semantic analysis phase of the compiler mainly checks for:					1
	a. Syntax errors	b. Type mismatches and scope errors	c. Efficiency of the program	d. Token generation		
10	Which of the following is a common type of grammar used in compiler construction?					1
	a. Context-free grammar	b. Regular grammar	c. Context-sensitive grammar	d. All of the above		
11	A compiler's code optimization phase aims to:					1
	a. Detect syntax errors	b. Minimize the runtime of the program	c. Translate high-level code to assembly	d. Create an abstract syntax tree		
12	The main task of a lexical analyzer is to:					1
	a. Parse the source code	b. Translate the high-level language to machine code	c. Break the source code into tokens	d. Optimize the generated code		
Q. 2	Solve the following.					12

A)	What are the differences between top-down and bottom-up parsing?		6
B)	Explain the working of LL(1) and LR(1) parsers?		6
Q.3	Solve the following.		12
A)	What are the primary responsibilities of the lexical analyzer?		6
B)	How are regular expressions used to specify tokens, and how do finite automata implement these specifications?		6
Q. 4	Solve Any Two of the following.		12
A)	What is the difference between a parse tree and an abstract syntax tree?		6
B)	What are the advantages and disadvantages of recursive descent parsers?		6
C)	How does shift-reduce parsing work?		6
Q.5	Solve Any Two of the following.		12
A)	What are common forms of intermediate representations (e.g., three-address code, abstract syntax trees, control flow graphs)?		6
B)	How are function calls and returns managed in the generated code?		6
C)	How do these optimizations improve the efficiency of the generated code?		6
Q. 6	Solve Any Two of the following.		12
A)	What are the differences between high-level and low-level optimizations?		6
B)	What is data flow analysis and how is it used in optimization?		6
C)	What are the primary goals of code optimization?		6
	*** End ***		